Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

## Agile Development Process and Devops Lab (ISWE406P )

# LAB FAT

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Slot: L47+L48

Question:

# ONLINE TICKET CANCELLATION SYSTEM

**Problem Statement:**
Ensure tickets are cancelled and refunded correctly.

**Tasks:**

1) Develop cancellation and refund logic
2) Write JUnit tests for refund scenarios
3) Implement CI/CD pipeline with automated testing
4) Deploy services on Docker and Kubernetes

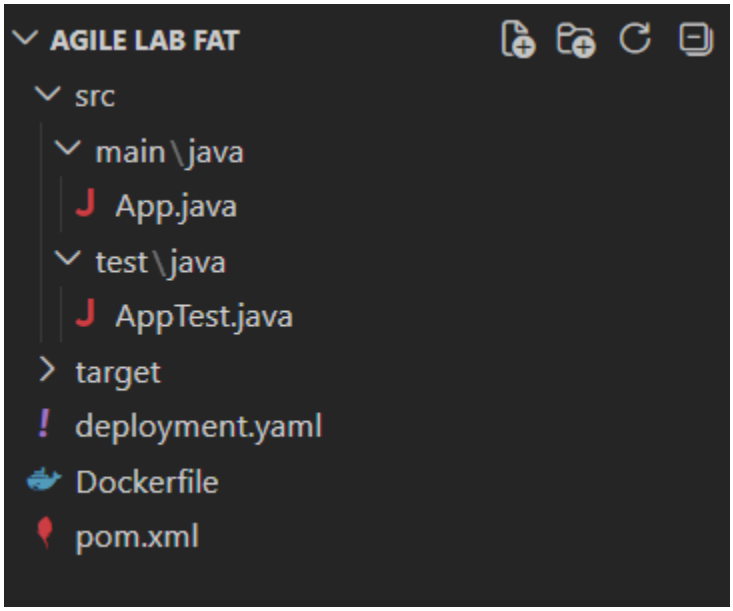
**Considerations while writing Java & JUnit code:**

- For Cancelling Ticket either Refund should be activated or payment should be unsuccessful

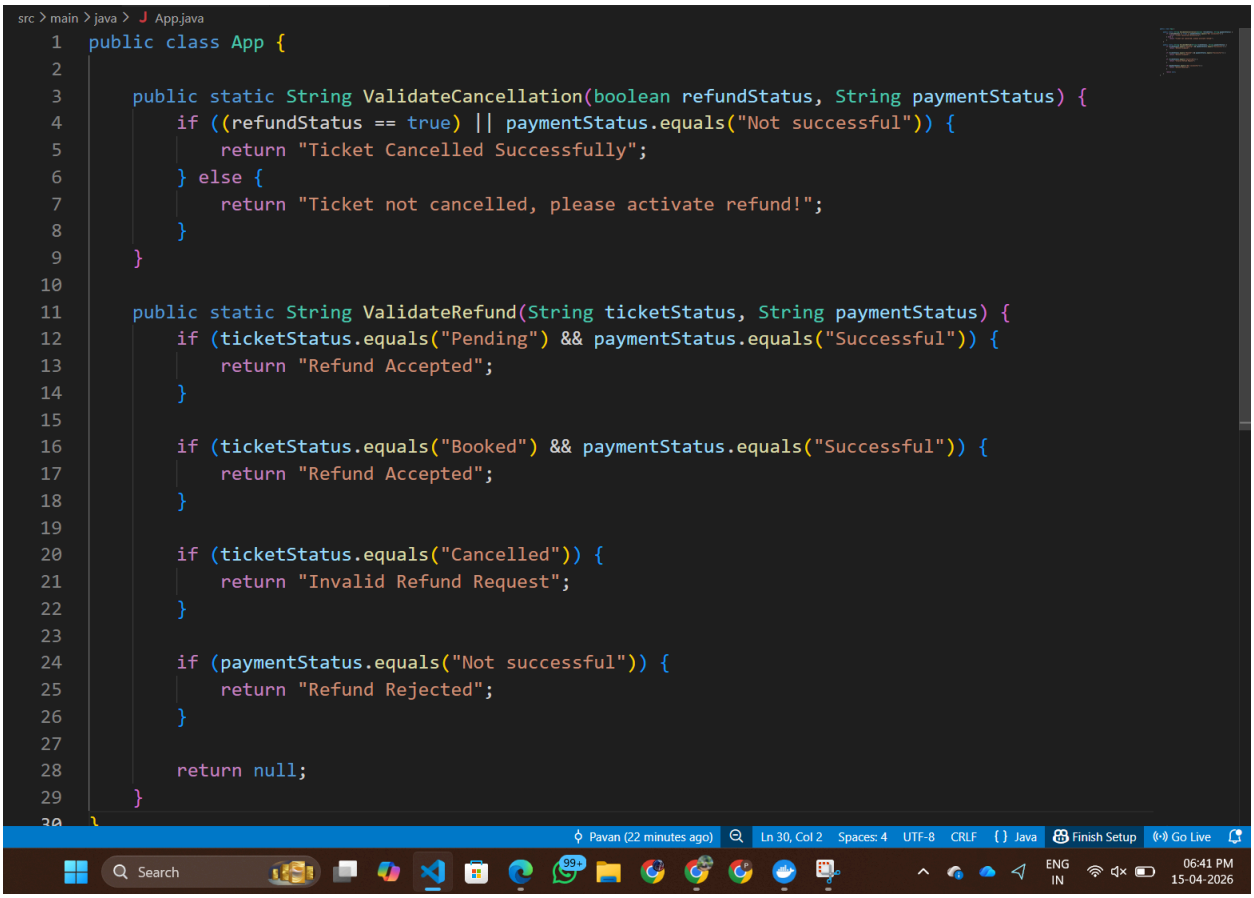
- For Refund Activation, the payment status has to be successful and ticket status can be either “Pending” or “Booked”

- If Ticket status is “Cancelled” and if someone has applied for Refund, it is considered as “Invalid Refund Request”

# Created Folder Structure:




# App.java:



src > main > java > App.java

```java
public class App {

    public static String ValidateCancellation(boolean refundStatus, String paymentStatus) {
        if ((refundStatus == true) || paymentStatus.equals("Not successful")) {
            return "Ticket Cancelled Successfully";
        } else {
            return "Ticket not cancelled, please activate refund!";
        }
    }

    public static String ValidateRefund(String ticketStatus, String paymentStatus) {
        if (ticketStatus.equals("Pending") && paymentStatus.equals("Successful")) {
            return "Refund Accepted";
        }

        if (ticketStatus.equals("Booked") && paymentStatus.equals("Successful")) {
            return "Refund Accepted";
        }

        if (ticketStatus.equals("Cancelled")) {
            return "Invalid Refund Request";
        }

        if (paymentStatus.equals("Not successful")) {
            return "Refund Rejected";
        }

        return null;
    }
}
```

# AppTest.java:

src > test > java > AppTest.java

```java
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;

public class AppTest {

    @Test
    void testValidateRefund() {
        assertEquals("Refund Accepted", App.ValidateRefund("Pending", "Successful"));

        assertEquals("Refund Rejected", App.ValidateRefund("Pending", "Not successful"));

        assertEquals("Refund Accepted", App.ValidateRefund("Booked", "Successful"));

        assertEquals("Refund Rejected", App.ValidateRefund("Booked", "Not successful"));

        assertEquals("Invalid Refund Request", App.ValidateRefund("Cancelled", "Successful"));

        assertEquals("Invalid Refund Request", App.ValidateRefund("Cancelled", "Not successful"));
    }
}
```

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# Pom.xml:

pom.xml

```xml

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.demo</groupId>
<artifactId>test-java-app</artifactId>
<version>1.0</version>
<packaging>jar</packaging>
<name>test-java-app</name>
<properties>
<maven.compiler.source>17</maven.compiler.source>
<maven.compiler.target>17</maven.compiler.target>
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
<junit.version>5.10.1</junit.version>
</properties>
<dependencies>
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<version>${junit.version}</version>
<scope>test</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
```

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# Deployment.yaml:

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simple-java-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: simple-java-app
  template:
    metadata:
      labels:
        app: simple-java-app
    spec:
      containers:
        - name: simple-java-app
          image: 23mis0563/java-test-app:latest
          ports:
            - containerPort: 8080

---
apiVersion: v1
kind: Service
metadata:
  name: java-service
spec:
  type: NodePort
  selector:
    app: simple-java-app
  ports:
```

# Dockerfile:

Dockerfile

```dockerfile
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

# Locally testing:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO]
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.369 s
[INFO] Finished at: 2026-04-15T18:15:50+05:30
[INFO] ------------------------------------------------------------------------

powershell
powershell
powershell

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```

# Pushed this directory into github:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/classes/App.class
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/classes/App.class
 create mode 100644 target/classes/App.class
 create mode 100644 target/classes/App.class
 create mode 100644 target/classes/App.class
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/AppTest.txt
 create mode 100644 target/surefire-reports/TEST-AppTest.xml
 create mode 100644 target/test-classes/AppTest.class
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
 create mode 100644 target/classes/App.class
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/AppTest.txt
 create mode 100644 target/surefire-reports/TEST-AppTest.xml
 create mode 100644 target/test-classes/AppTest.class
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
 create mode 100644 target/surefire-reports/TEST-AppTest.xml
 create mode 100644 target/test-classes/AppTest.class
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
PS C:\Users\pavan\Documents\AGILE LAB FAT> git branch -M main
PS C:\Users\pavan\Documents\AGILE LAB FAT> git remote add origin https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT
PS C:\Users\pavan\Documents\AGILE LAB FAT> git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 16 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (30/30), 5.73 KiB | 279.00 KiB/s, done.
Total 30 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\pavan\Documents\AGILE LAB FAT>

powershell
powershell
powershell

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```
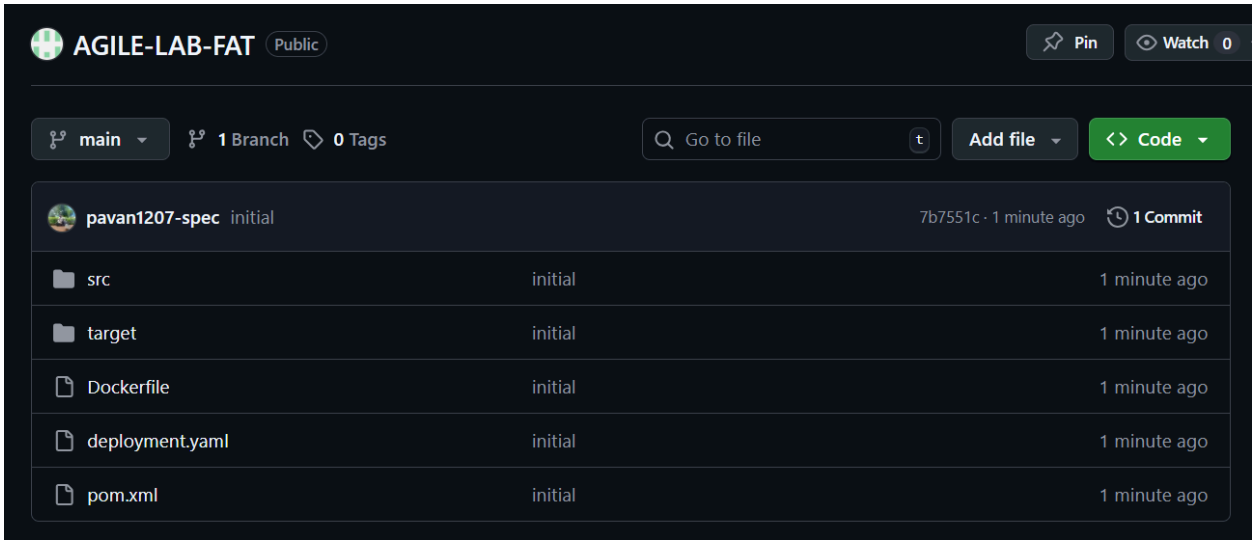
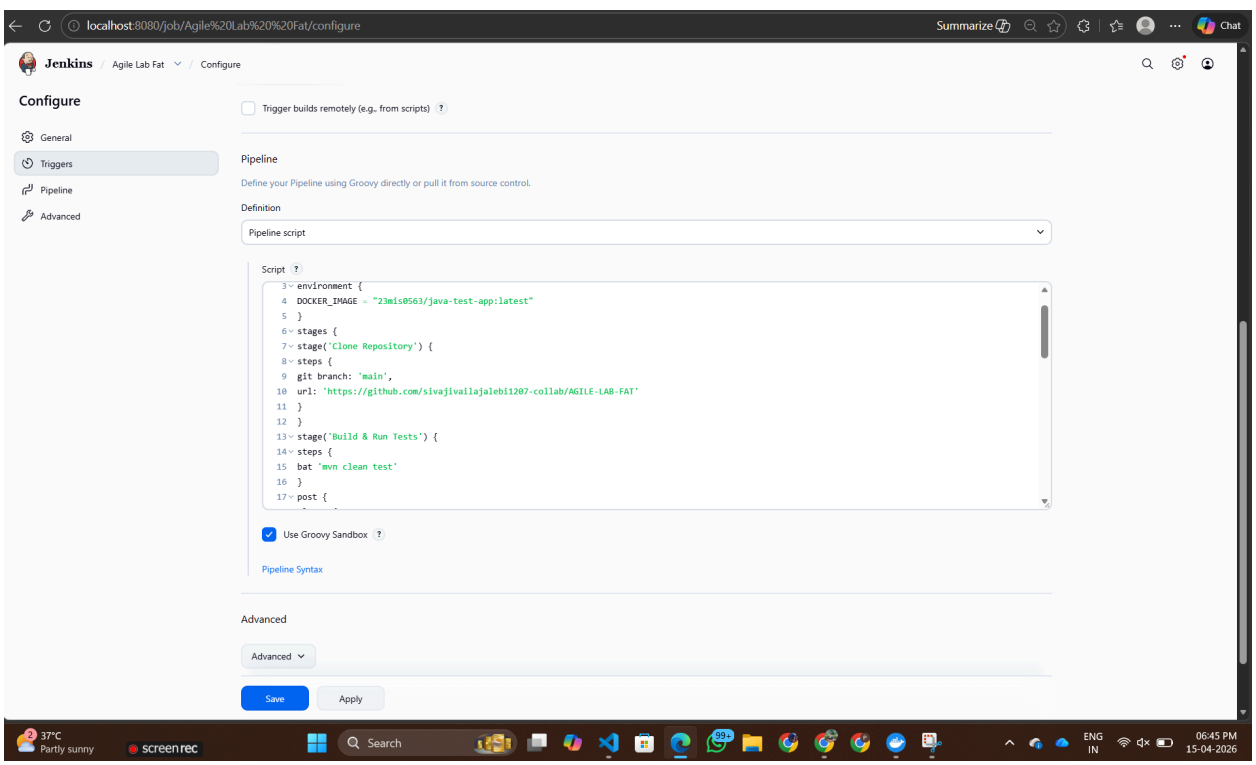
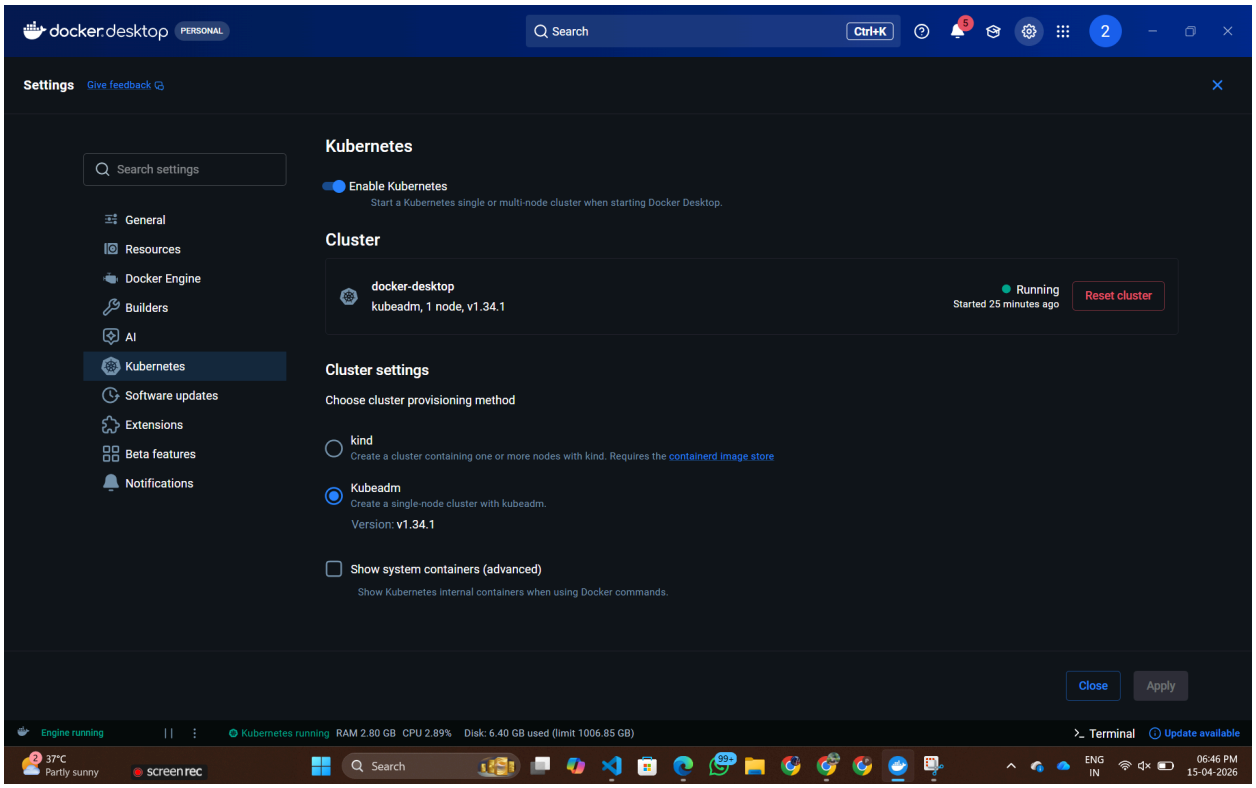



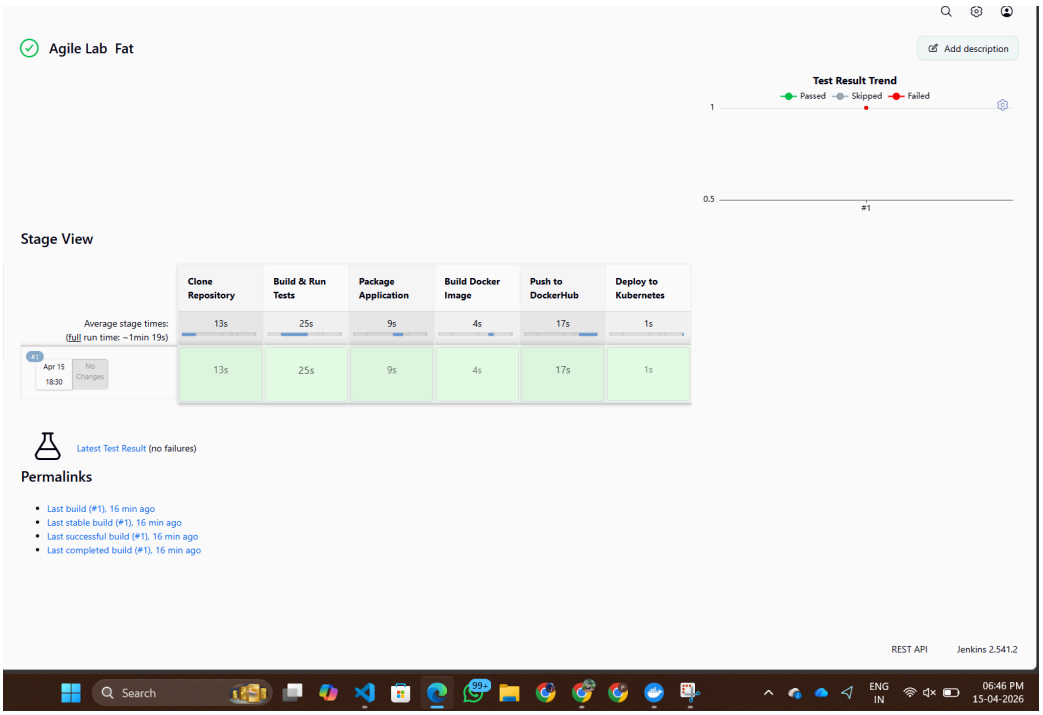

Created a new Pipeline Item in Jenkins and added a pipeline script:



Enabled Kubernetes on Docker Desktop:




Created a new Build in Jenkins and the Build was successful:



# Console Output in Jenkins:

Console

Download | Copy | View as plain text

```
Started by user Pavan Kalyan K
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\Agile Lab  Fat
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repository)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\Agile Lab  Fat # timeout=10
Fetching upstream changes from https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT
 > git.exe --version # timeout=10
 > git --version # 'git version 2.47.1.windows.2'
 > git.exe fetch --tags --force --progress -- https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT +refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe config remote.origin.url https://github.com/sivajivailajalebi1207-collab/AGILE-LAB-FAT # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 7b7551c47a863a9d6c0c3ac8a43de678fe3a0801 (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 7b7551c47a863a9d6c0c3ac8a43de678fe3a0801 # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe checkout -b main 7b7551c47a863a9d6c0c3ac8a43de678fe3a0801 # timeout=10
Commit message: "initial"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build & Run Tests)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Agile Lab  Fat>mvn clean test
[INFO] Scanning for projects...
```

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## ➔ Junit Test by Maven:

```
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[INFO] skip non existing resourceDirectory C:\ProgramData\Jenkins\.jenkins\workspace\Agile Lab  Fat\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[WARNING] system modules path not set in conjunction with -source 17
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.101 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  15.451 s
[INFO] Finished at: 2026-04-15T18:31:20+05:30
[INFO] ------------------------------------------------------------------------
Post stage
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Package Application)
[Pipeline] bat
```

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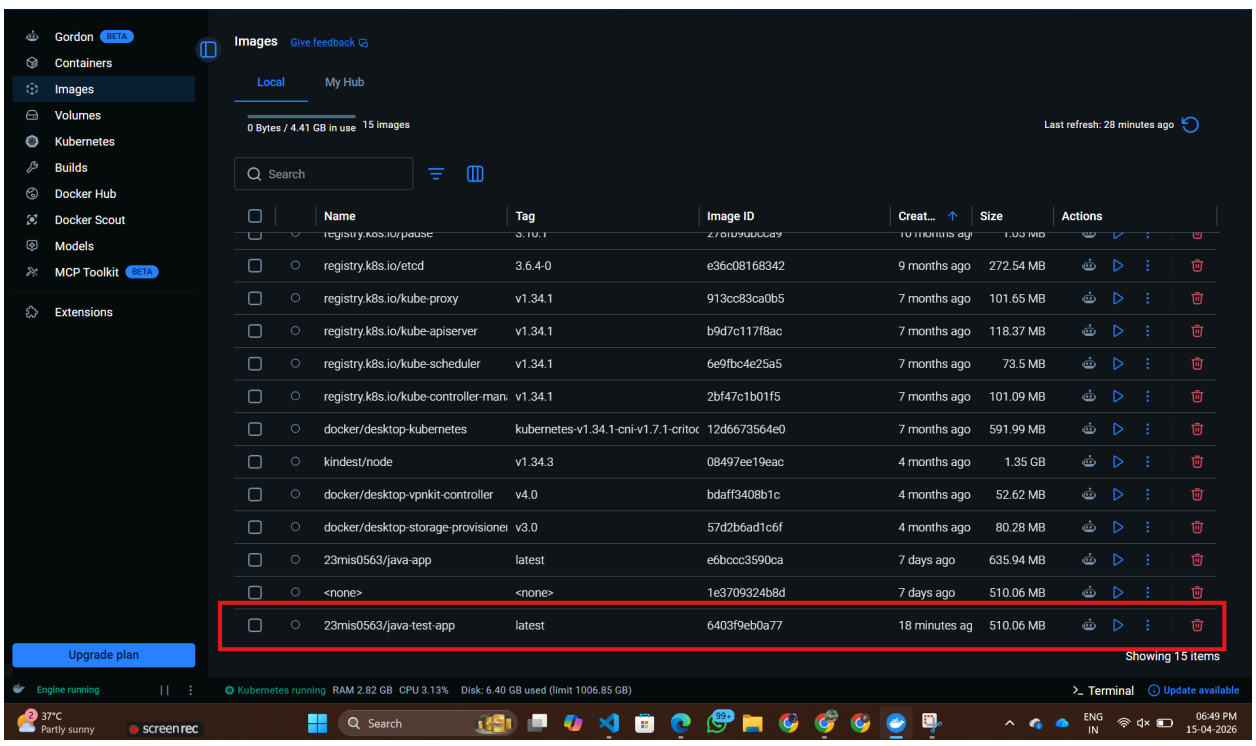
## ➔ Pipeline Executed Successfully:

```
82c540078c28: Layer already exists
946c51efe0fd: Layer already exists
13b9be02f27d: Waiting
13b9be02f27d: Waiting
943851fc7b79: Pushed
13b9be02f27d: Pushed
latest: digest: sha256:6403f9eb0a776856267daaaed32fc9625c86533e4e37f0c8622136b6fcb20457 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Agile Lab  Fat>set KUBECONFIG=C:/ProgramData/Jenkins/.jenkins/workspace/new_practice/.kube/config

C:\ProgramData\Jenkins\.jenkins\workspace\Agile Lab  Fat>kubectl apply -f deployment.yaml
deployment.apps/simple-java-app unchanged
service/java-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.2

# Deployed and Image Created on Docker Desktop: